

Introduction

Lycorma delicatula (Spotted Lanternfly) is native to China and Southeast Asia. It was first found in the United States, in Pennsylvania, in 2014.



Figure 1. *Lycorma delicatula*

The spotted lanternfly has no native predators in the United States and prefers to feed on the *Ailanthus altissima* (tree of heaven), which is also invasive and native to China and some parts of Vietnam. The *Lycorma delicatula* also feeds on some native vegetation, damaging vineyards in Loudoun County. According to a Congressional Report in 2023, *Lycorma delicatula* costs over \$324 million annually and cost more than 2,800 jobs to Pennsylvania alone.

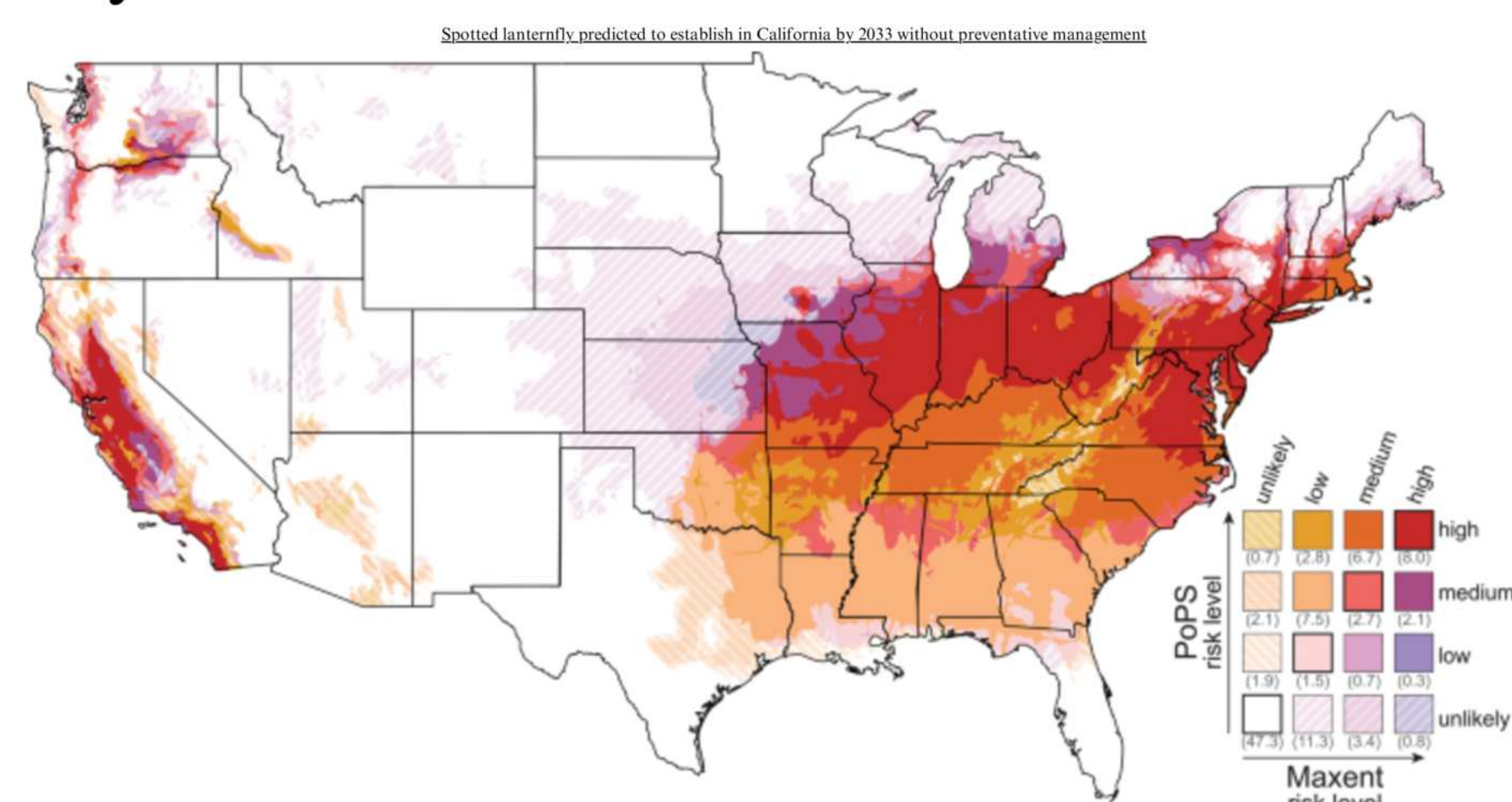


Figure 2. Predicted Spread of *Lycorma delicatula*

Purpose

The purpose of this study is to investigate whether different lipase enzymes can break down the protective waxy ootheca coating of *Lycorma delicatula* eggs to evaluate a potential biochemical method for controlling this invasive species.

- The information from this research will help provide options to **counter and control this pest**, such as developing attractants to lure the pests or developing better pesticides.
- While the exact chemical composition of the ootheca is **not fully known**, this experiment hypothesizes that waxy ester compounds are present and can be degraded by lipase enzymes.
- By comparing different lipase sources, this experiment aims to identify which enzymes are most effective at **degrading the ootheca coating**.

Targeting *Lycorma delicatula* Ootheca Using *Candida rugosa* and Goat Lipase Enzymes in Leesburg, Virginia

Environmental Chemistry

Methods and Materials

Preparation

- Collect samples (Powell et al., 2023)
- Prepare Tris-HCl buffer for enzyme activity (MilliporeSigma, n.d.)

Chemical Analysis

- Dissolve lipase enzymes in deionized water at a standardized concentration (MilliporeSigma, n.d.)
- Use FTIR to analyze ootheca chemistry before and after treatment (Powell et al., 2023)
- Record visual changes and mass

Data Analysis

- Compare control vs. treated samples
- Analyze results using FTIR analysis

Figure 3. SLF Egg Case



Figure 4. Ootheca Submerged in Lipase

Chemicals Used:

- Lipase from *Candida rugosa* (fungus)
- 100% Goat Lipase (sharp)

Collection

Egg cases of *Lycorma delicatula* will be collected around Leesburg, Virginia, in mid-to-late November and carefully removed using a spatula or knife. Samples will be stored individually in paper envelopes until analysis.



Figure 5. *Ailanthus altissima* Sample Collection

Results

Preliminary FTIR analysis of *Lycorma delicatula* ootheca and day 5 and day 20 lipase exposure tests.

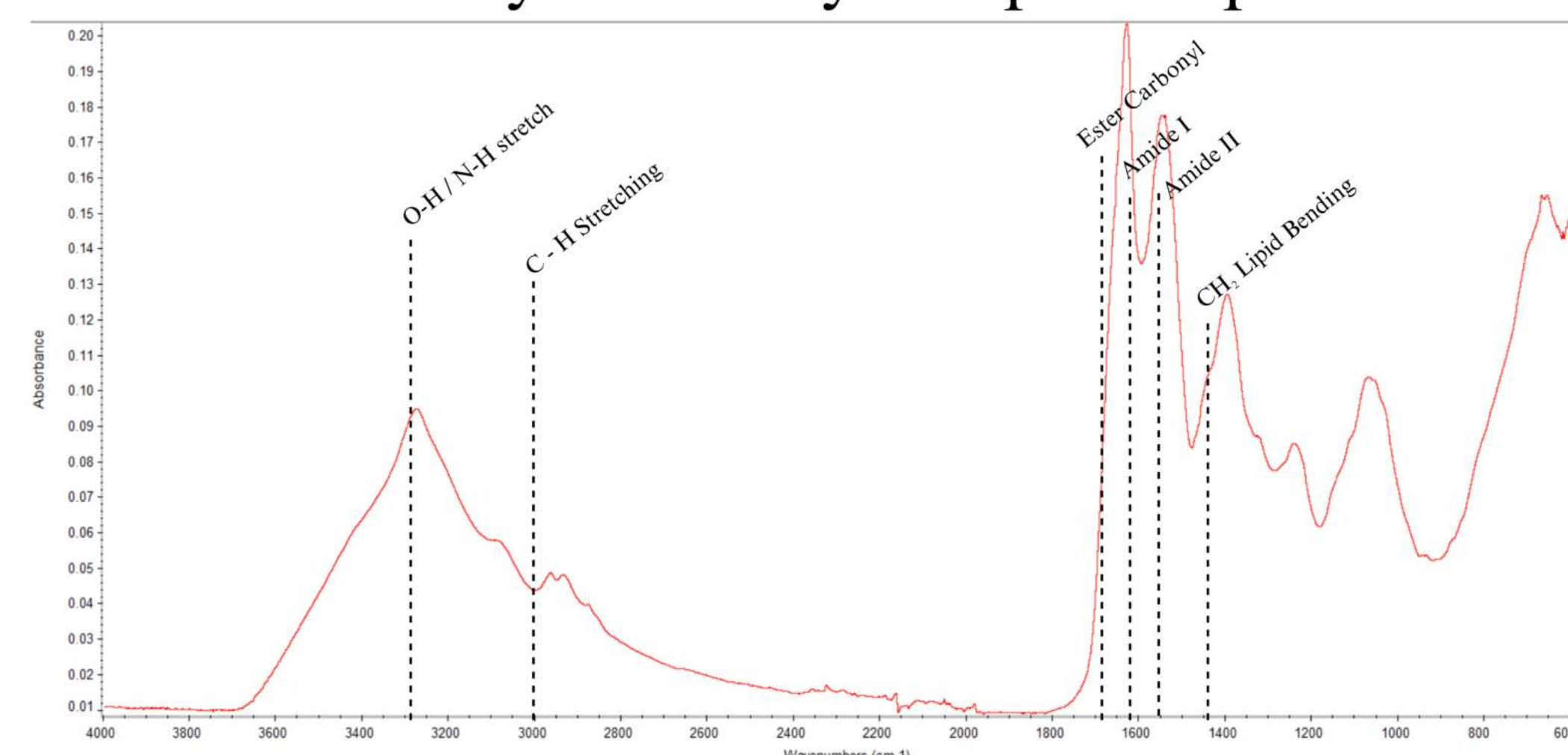


Figure 6. SLF Ootheca Control Test

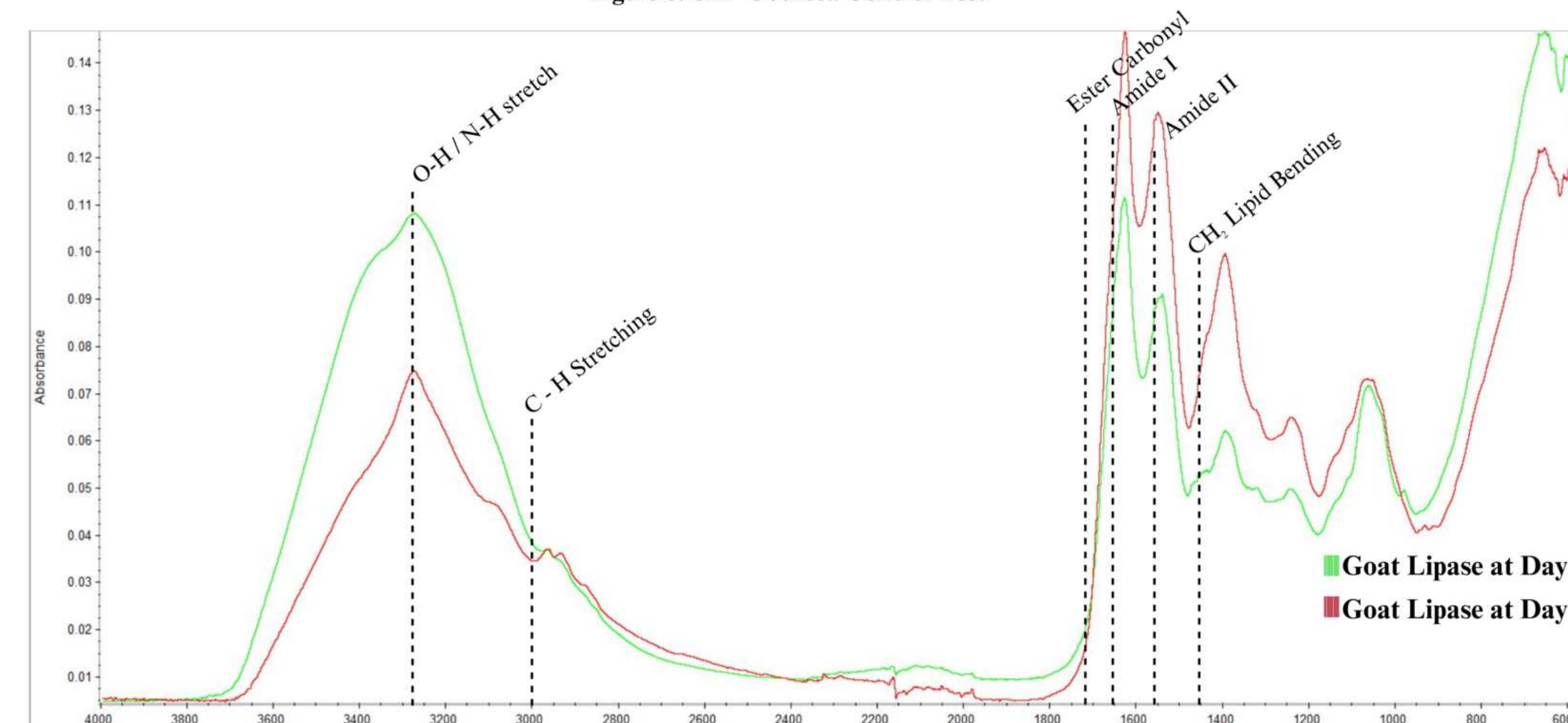


Figure 7. SLF Ootheca Goat Lipase Enzyme Test at Day 5 and Day 20

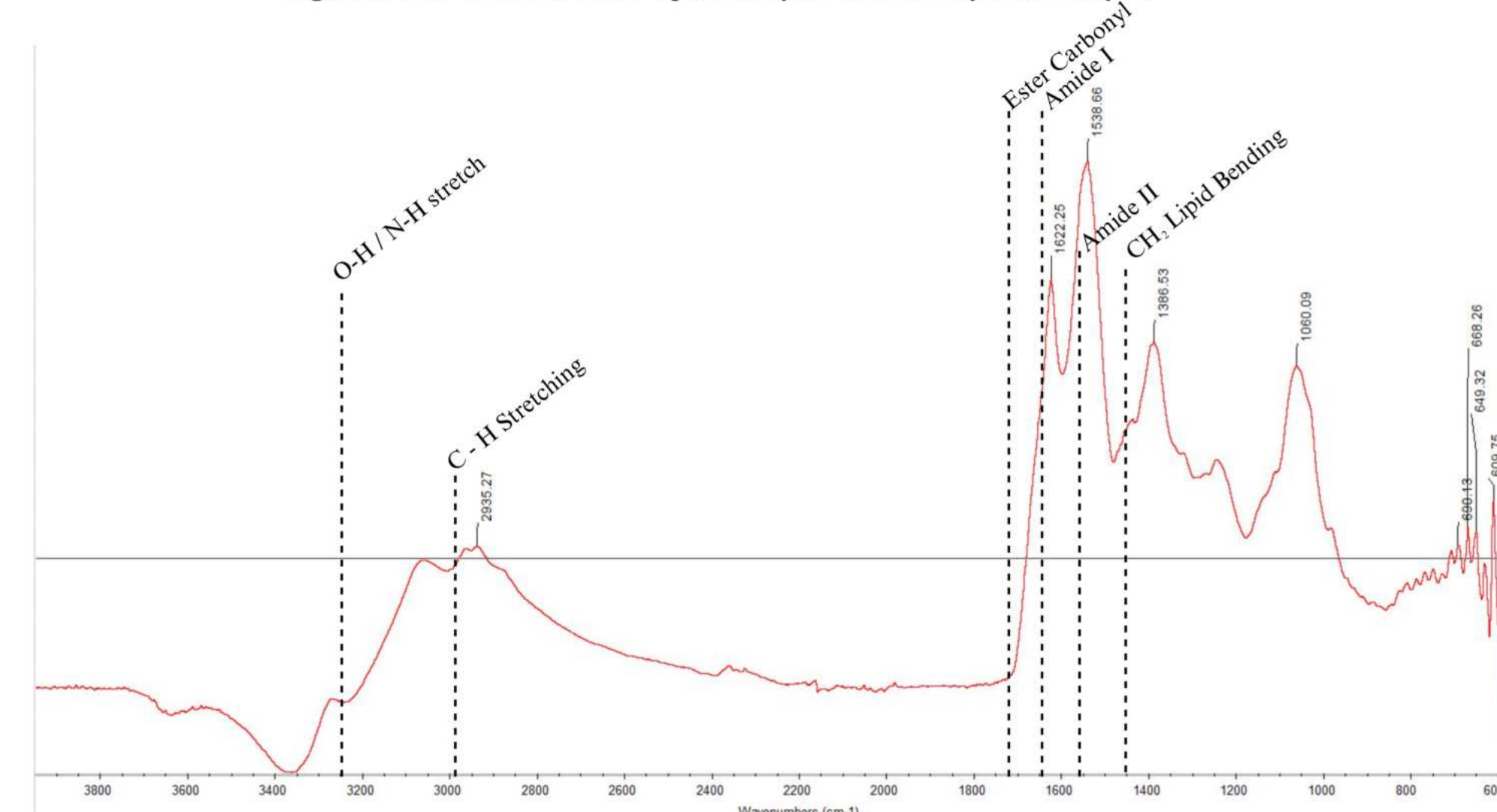


Figure 8. Day 5 SLF Ootheca *Candida rugosa* Lipase Enzyme Test

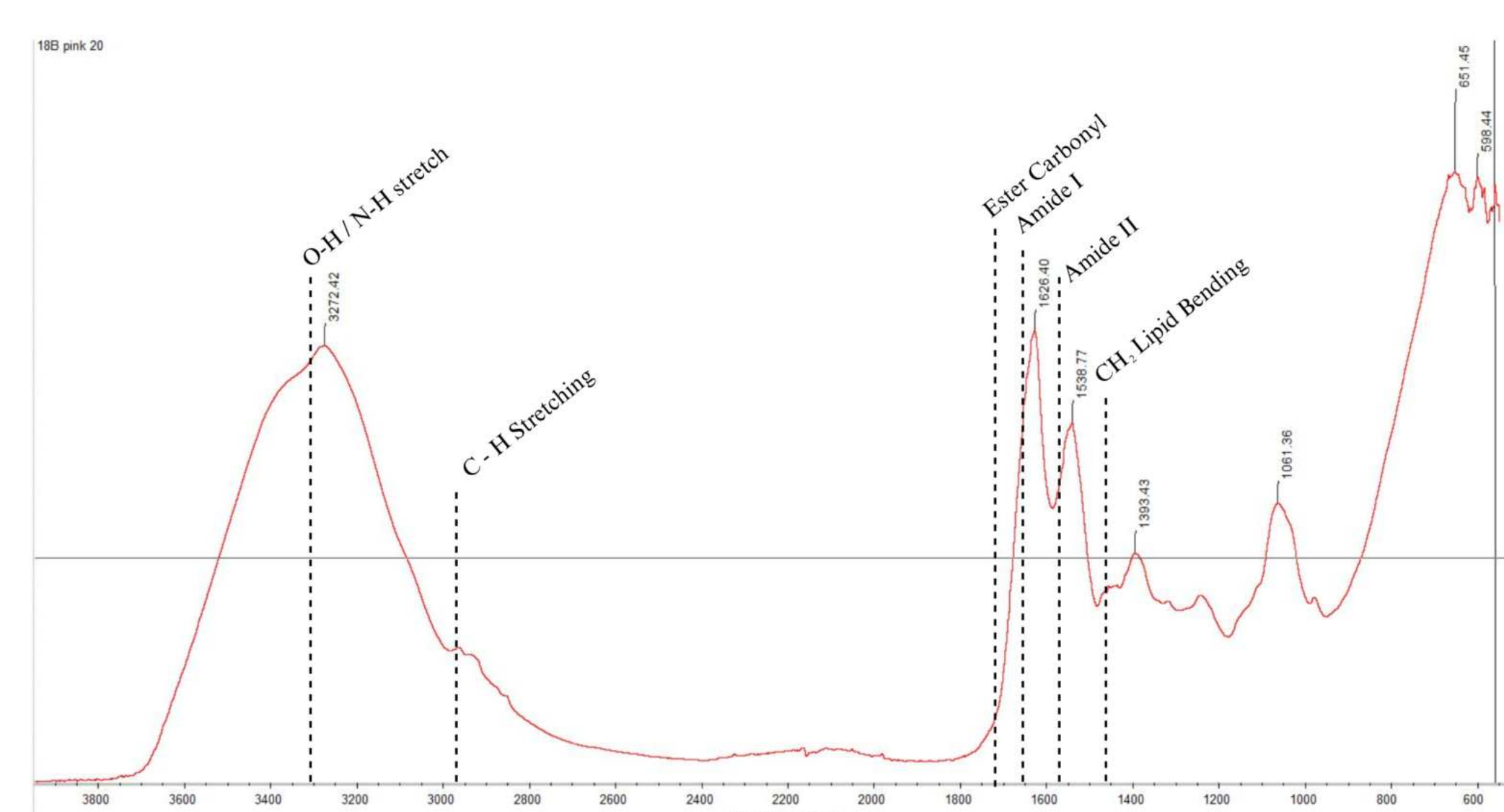


Figure 9. Day 20 SLF Ootheca *Candida rugosa* Lipase Enzyme Test

Conclusions

Based on the baseline chemical and structural analysis of the *Lycorma delicatula* ootheca, several key observations and potential outcomes for enzymatic treatment can be drawn.

- Preliminary FTIR analysis of *Lycorma delicatula* ootheca shows a **protein-rich, highly structured protective matrix**.
- Strong amide peaks indicate **chemical similarity to keratin-rich materials** like human hair, which explains the durability and pesticide resistance of the egg cases.
- Hydroxyl and weak ester signals suggest that **ester-linked waxes** may be present, embedded within the protein matrix.

Chemical changes following enzymatic treatment:

- Day 20 samples show stronger 3000cm⁻¹ peaks, indicating increased **exposure of hydrophobic groups C-H** as the matrix breaks down.
- Amide I (~1600cm⁻¹) decreases in treated samples, **supporting protein-matrix disruption**.
- 1400cm⁻¹ decreases suggest changes to **lipid-linked or side-chain structures**, consistent with enzymatic modification of waxy or ester components
- Hydrogen bonding peak at 3200cm⁻¹ shrinks at Day5 but grows at Day 20, indicating **early disruption** followed by **increased hydration** as it becomes more porous.

Future Research

- Using this information, I will **validate my results** with increased samples and lipase variation
 - Future research will focus on optimizing enzyme concentration, exposure time, and delivery methods, as well as exploring other biochemical targets in the ootheca to improve effectiveness for **real-world SLF management**.

References

Unless specifically indicated, all images, figures, and tables used in this presentation were created by the researcher.
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